AI Agents

Using to Improve Design Output

Introduction

Objective

Explain how purpose built AI agents are used across the design life cycle to improve speed, clarity, and quality of design outputs.

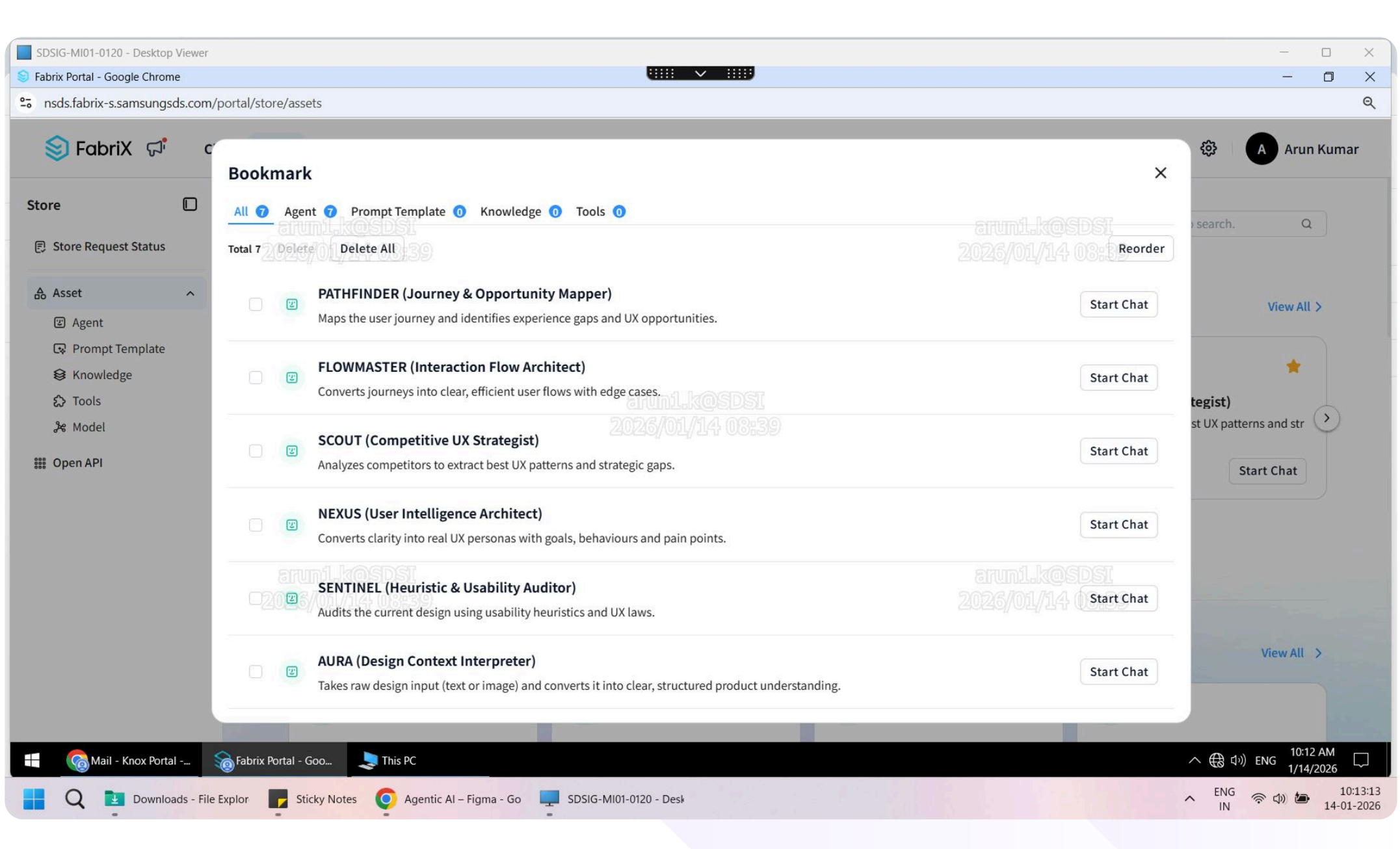
Key Message

Instead of one generic AI, we use specialized agents, each trained and prompted for a specific design task.

Why AI Agents

- Design work is multi-dimensional: research, strategy, flows, validation, and synthesis
- One prompt → one agent → one clear outcome
- Faster results and improves consistency

Our AI Agent Ecosystem



Bookmark

All · Agent · Prompt Template · Knowledge · Tools

PATHFINDER (Journey & Opportunity Mapper) – Maps the user journey and identifies experience gaps and UX opportunities.

FLOWMASTER (Interaction Flow Architect) – Converts journeys into clear, efficient user flows with edge cases.

SCOUT (Competitive UX Strategist) – Analyzes competitors to extract best UX patterns and strategic gaps.

NEXUS (User Intelligence Architect) – Converts clarity into real UX personas with goals, behaviours and pain points.

SENTINEL (Heuristic & Usability Auditor) – Audits the current design using usability heuristics and UX laws.

AURA (Design Context Interpreter) – Takes raw design input (text or image) and converts it into clear, structured product understanding.

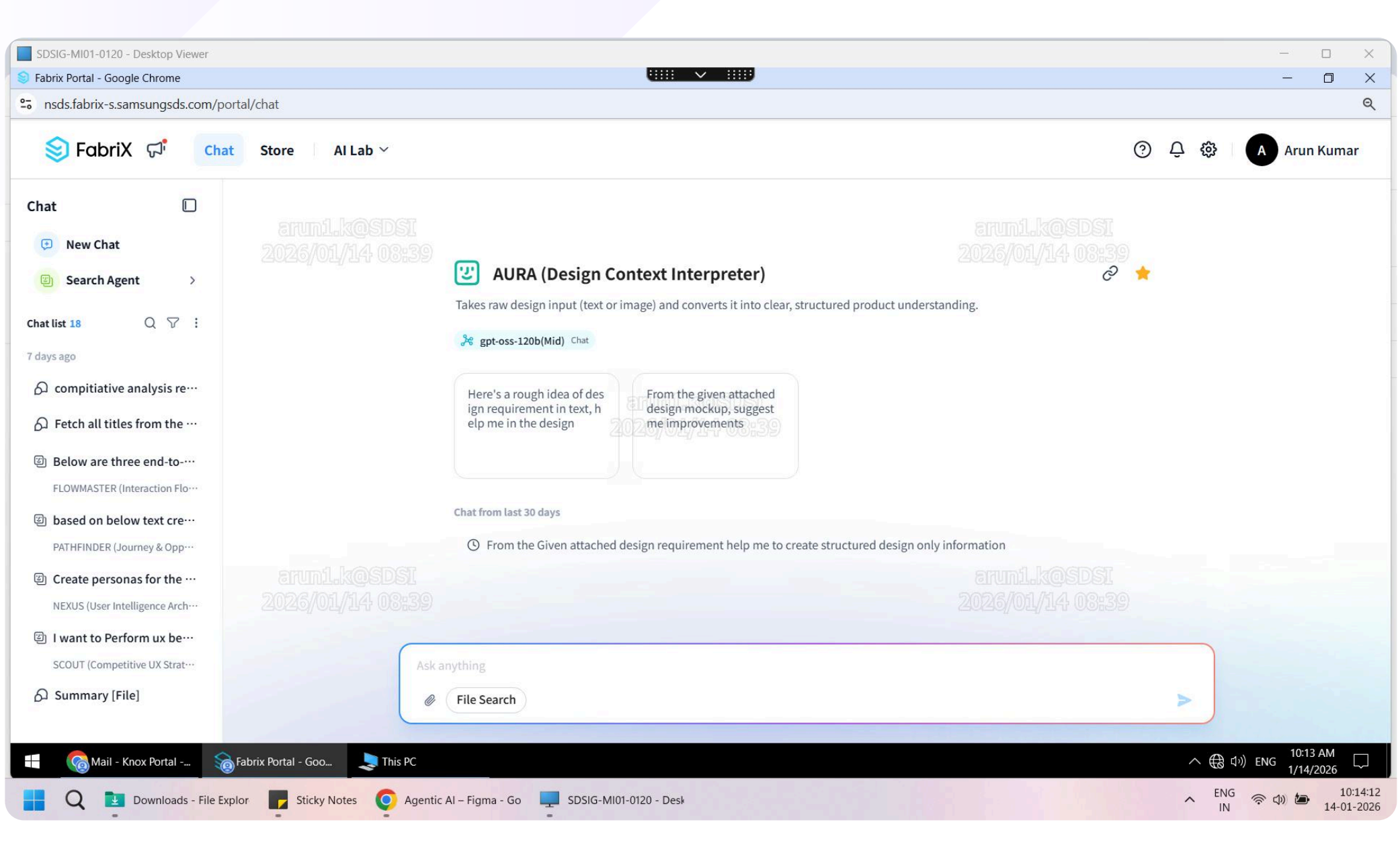
Agent Name	Role	Design Stage
1. Aura	Design Context Interpreter	Requirement Understanding
2. Pathfinder	Journey & Opportunity Mapper	Discovery & UX Strategy
3. Nexus	User Intelligence Architect	Empathy & Personas
4. Journey Mapper	User Journey & Oppurtunity	User Journey Map
5. Flow-master	Interaction Flow Architect	IA & User Flows
6. Scout	Competitive UX Strategist	Benchmarking
7. Sentinel	Heuristic & Usability Auditor	Validation & QA

How the System Works

Input → Agent → Structured Output → Designer Action

1. Designer selects the right agent
2. Provides a focused prompt (Text, PRD PDF, DOC or Image)
3. Agent interprets input using its defined role
4. Output is structured, reusable, and design-ready

Example



AURA (Design Context Interpreter)

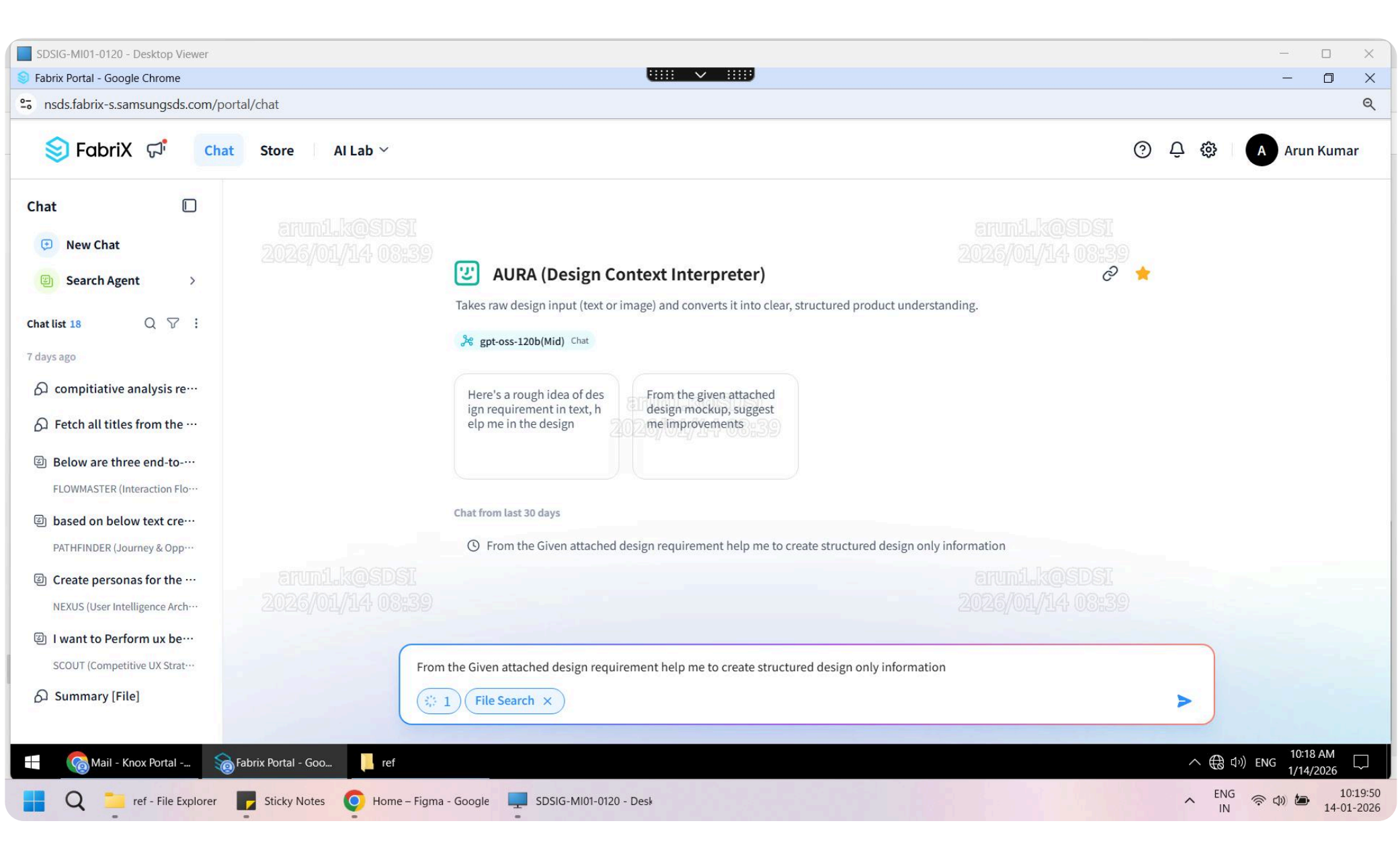
Takes raw design input (text or image) and converts it into clear, structured product understanding.

Here's a rough idea of design requirement in text, help me in the design

From the given attached design mockup, suggest me improvements

From the Given attached design requirement help me to create structured design only information

Attach file & Write prompt

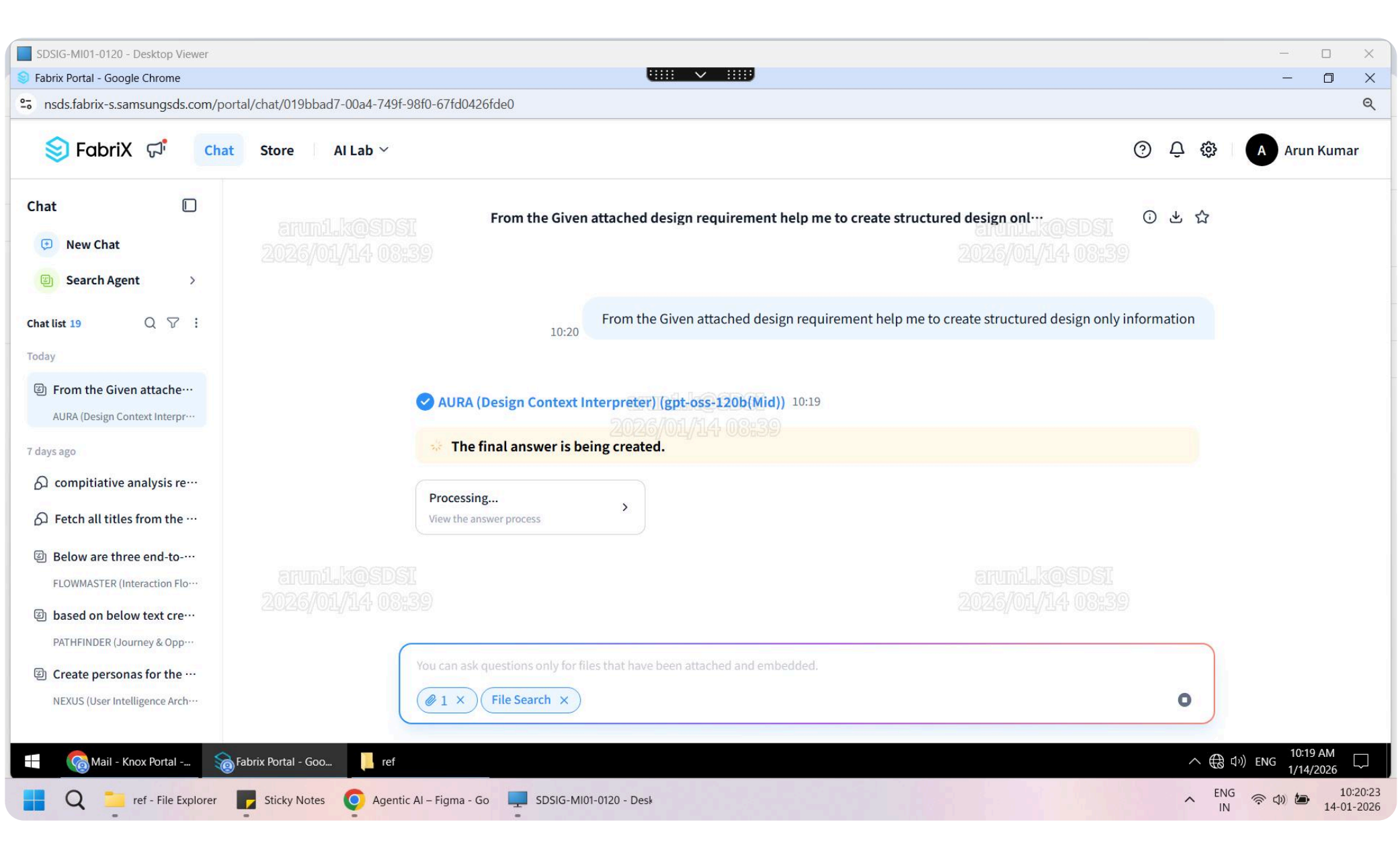


AURA (Design Context Interpreter)

Takes raw design input (text or image) and converts it into clear, structured product understanding.

From the Given attached design requirement help me to create structured design only information

Agent is analyzing the uploaded data & prompt before answering



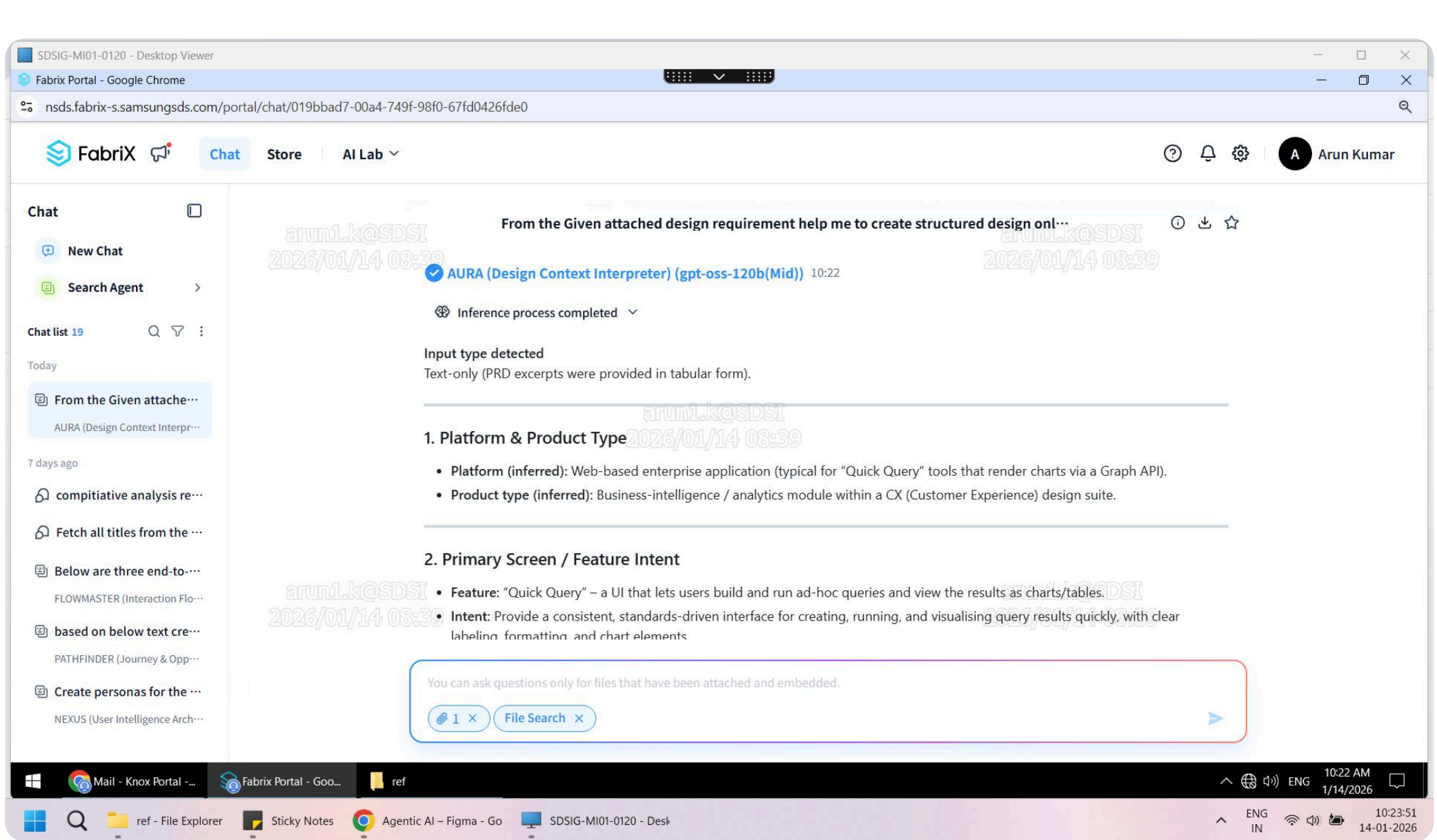
From the Given attached design requirement help me to create structured design only information

AURA (Design Context Interpreter) (gpt-oss-120b(Mid)) 10:19

The final answer is being created.

Processing...

Final Result



From the Given attached design requirement help me to create structured design onl…

AURA (Design Context Interpreter) (gpt-oss-120b(Mid)) 10:21

Inference process completed

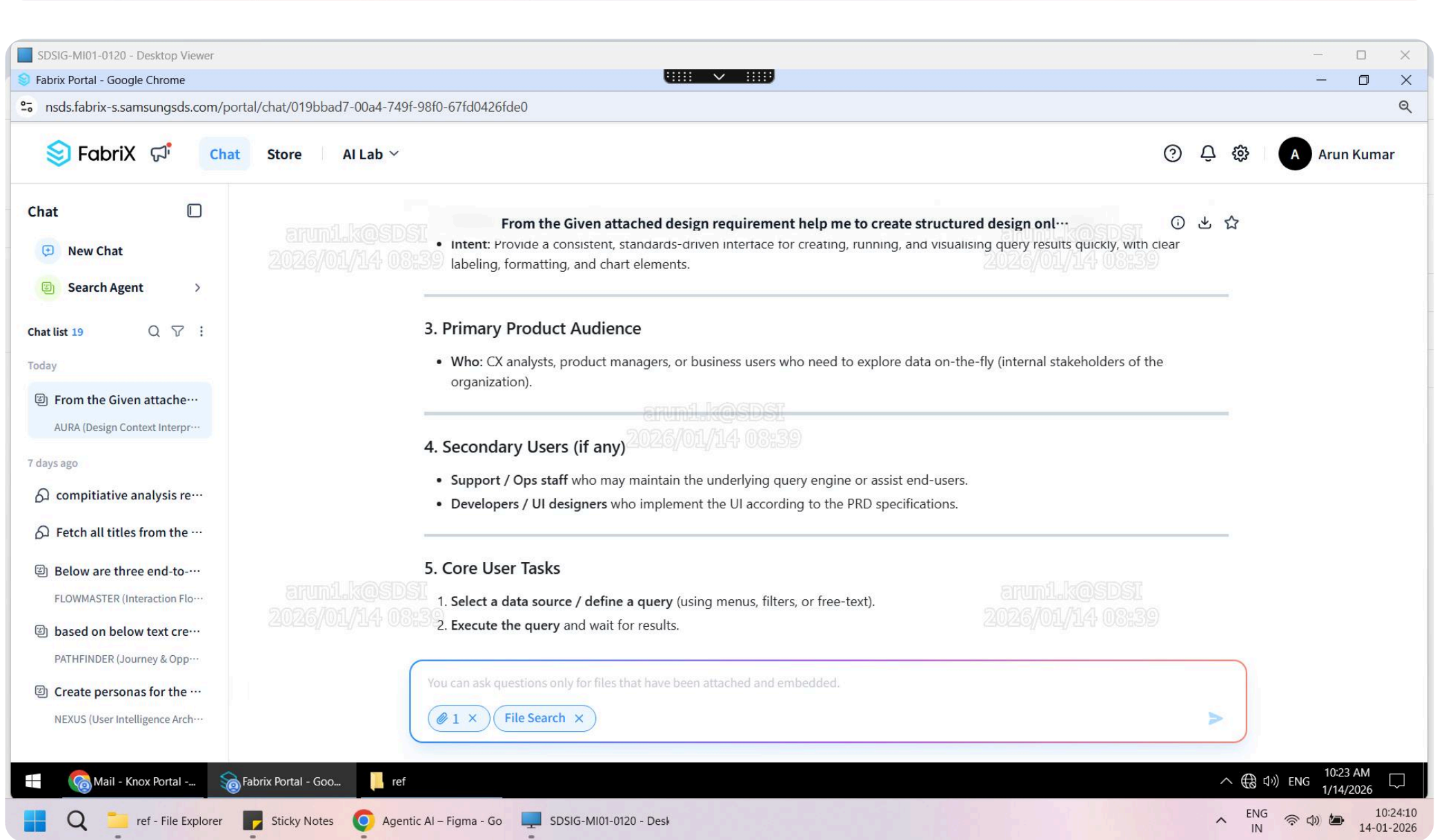
Input type detected
Text-only (PRD excerpts were provided in tabular form).

1. Platform & Product Type

- **Platform (inferred):** Web-based enterprise application (typical for "Quick Query" tools that render charts via a Graph API).
- **Product type (inferred):** Business intelligence / analytics module within a CX (Customer Experience) design suite.

2. Primary Screen / Feature Intent

- **Feature:** "Quick Query" – a UI that lets users build and run ad-hoc queries and view the results as charts/tables.
- **Intent:** Provide a consistent, standards-driven interface for creating, running, and visualizing query results quickly, with clear labeling, formatting, and chart elements.



3. Primary Product Audience

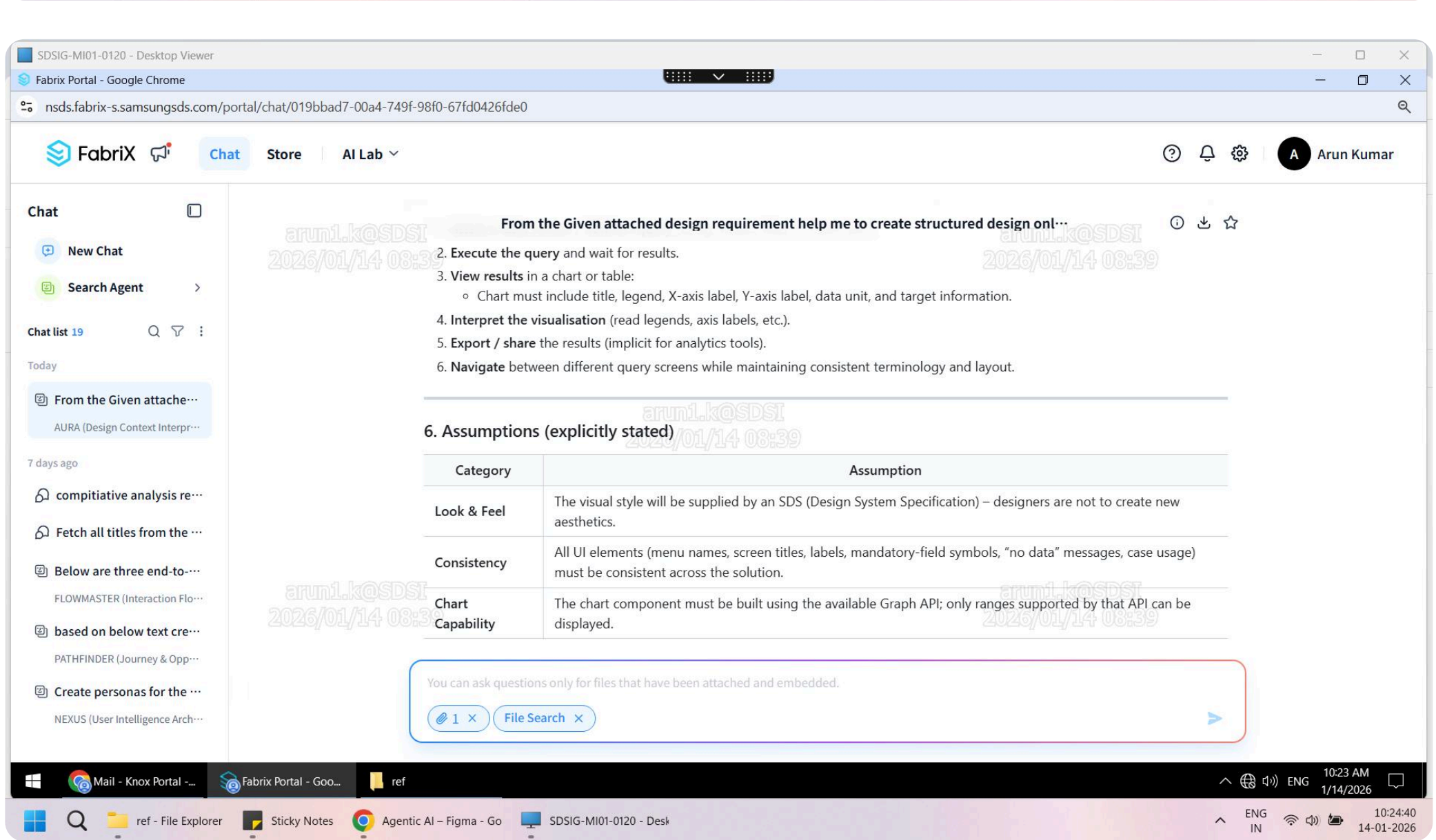
- **Who:** CX analysts, product managers, or business users who need to explore data on-the-fly (internal stakeholders of the organization).

4. Secondary Users (if any)

- **Support / Ops staff** who may maintain the underlying query engine or assist end users.
- **Developers / UI designers** who implement the UI according to the PRD specifications.

5. Core User Tasks

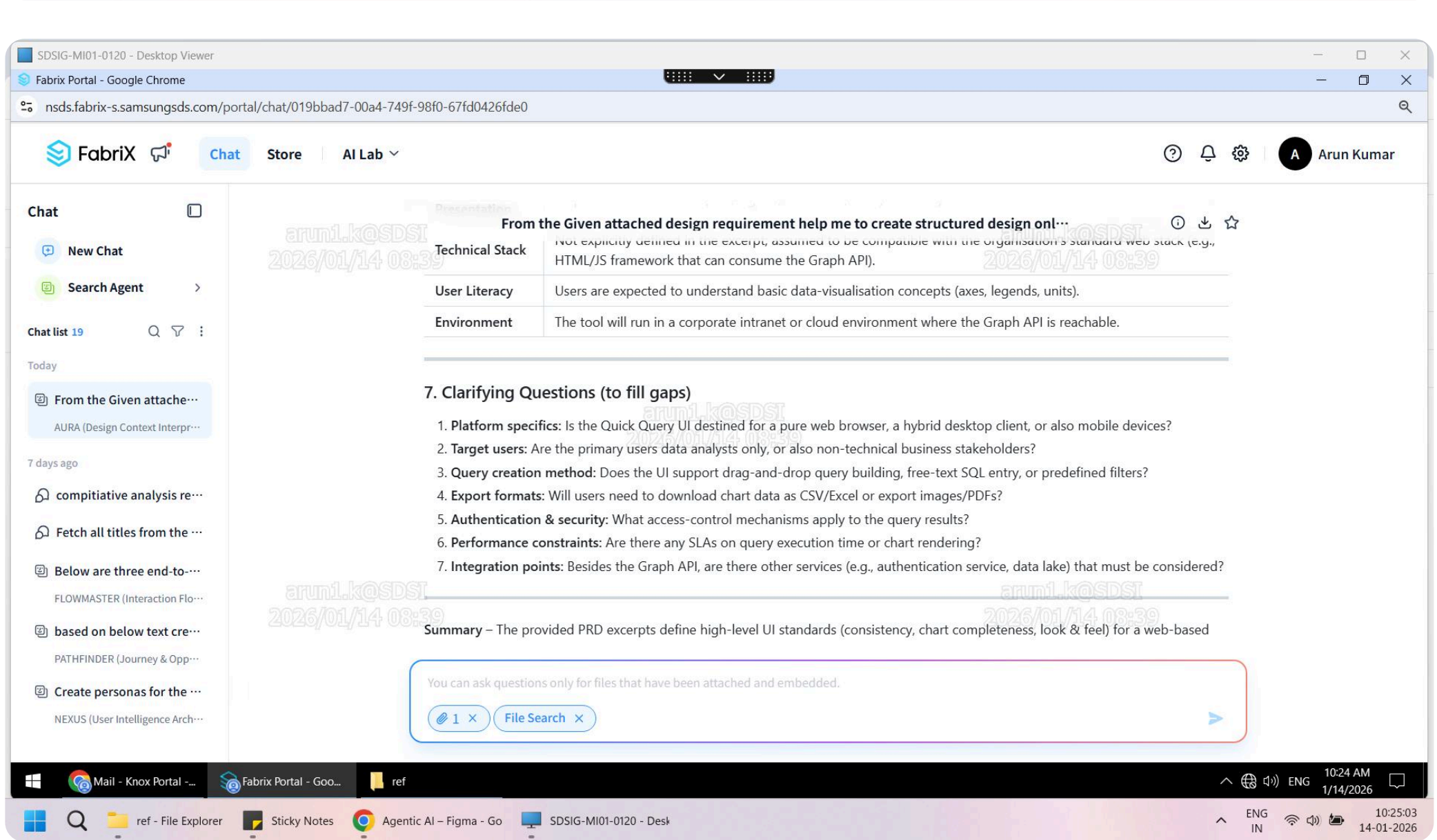
1. Select a data source / define a query (using menus, filters, or free-text).
2. Execute the query and wait for results.



2. Execute the query and wait for results.
3. View results in a chart or table:
   - Chart must include title, legend, X-axis label, Y-axis label, data unit, and target information.
4. Interpret the visualization (read legends, axis labels, etc.).
5. Export / share the results (implicit for analytics tools).
6. Navigate between different query screens while maintaining consistent terminology and layout.

6. Assumptions (explicitly stated)

Category	Assumption
Look & Feel	The visual style will be supplied by an SDS (Design System Specification) – designers are not to create new aesthetics.
Consistency	All UI elements (menu names, screen titles, labels, mandatory-field symbols, "no data" messages, case usage) must be consistent across the solution.
Chart Capability	The chart component must be built using the available Graph API; only ranges supported by that API can be displayed.



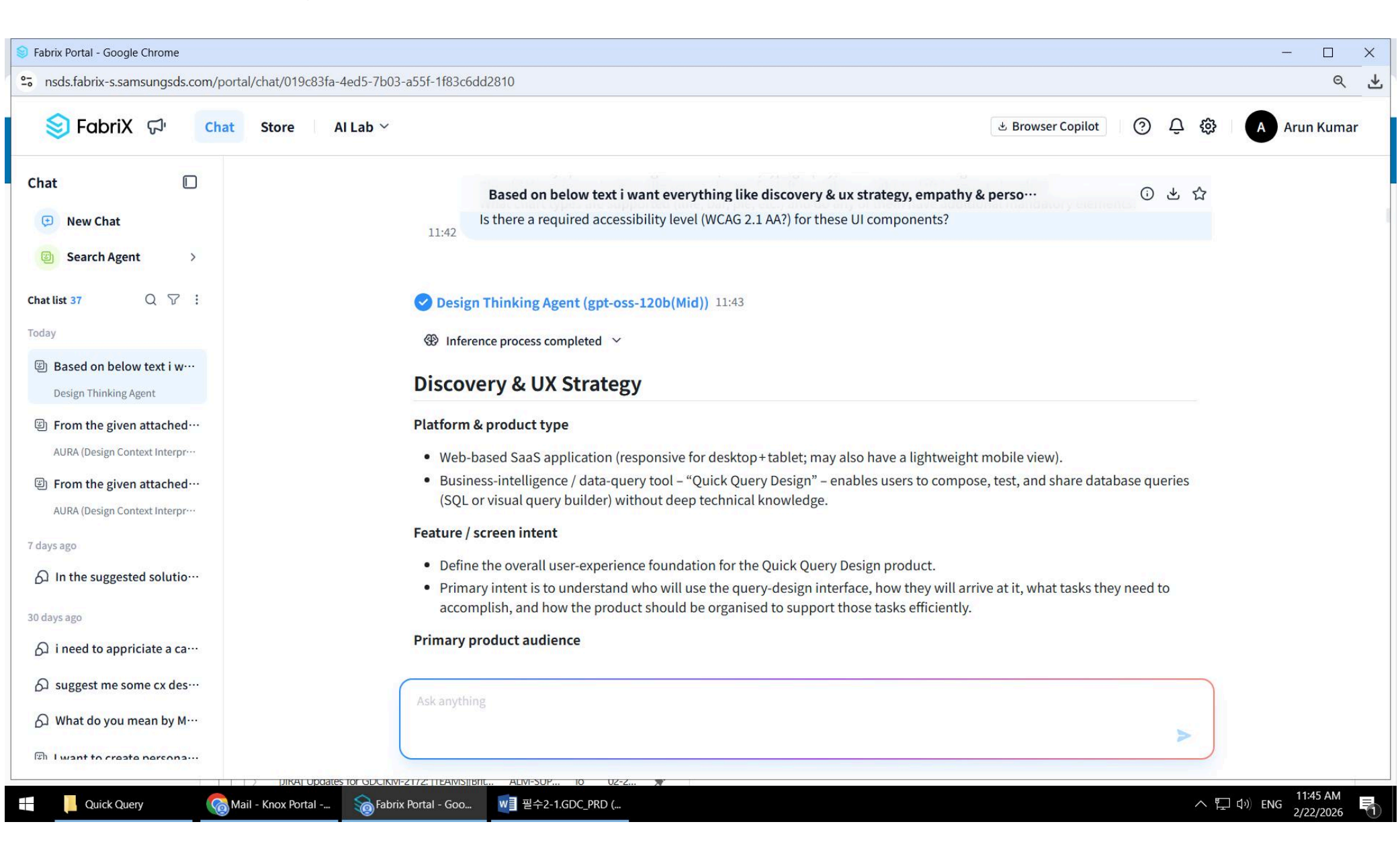
Technical Stack	… HTML/JS framework that can consume the Graph API).
User Literacy	Users are expected to understand basic data-visualization concepts (axes, legends, units).
Environment	The tool will run in a corporate intranet or cloud environment where the Graph API is reachable.

7. Clarifying Questions (to fill gaps)

1. **Platform specifics:** Is the Quick Query UI destined for a pure web browser, a hybrid desktop client, or also mobile devices?
2. **Target users:** Are the primary users data analysts only, or also non-technical business stakeholders?
3. **Query creation method:** Does the UI support drag-and-drop query building, free-text SQL entry, or predefined filters?
4. **Export formats:** Will users need to download chart data as CSV/Excel or export images/PDFs?
5. **Authentication & security:** What access-control mechanisms apply to the query results?
6. **Performance constraints:** Are there any SLAs on query execution time or chart rendering?
7. **Integration points:** Besides the Graph API, are there other services (e.g., authentication service, data lake) that must be considered?

Summary – The provided PRD excerpts define high-level UI standards (consistency, chart completeness, look & feel) for a web-based

Discovery & UX Strategy



Based on below text i want everything like discovery & ux strategy, empathy & perso…

Is there a required accessibility level (WCAG 2.1 AA?) for these UI components?

Design Thinking Agent (gpt-oss-120b(Mid)) 11:45

Inference process completed

Discovery & UX Strategy

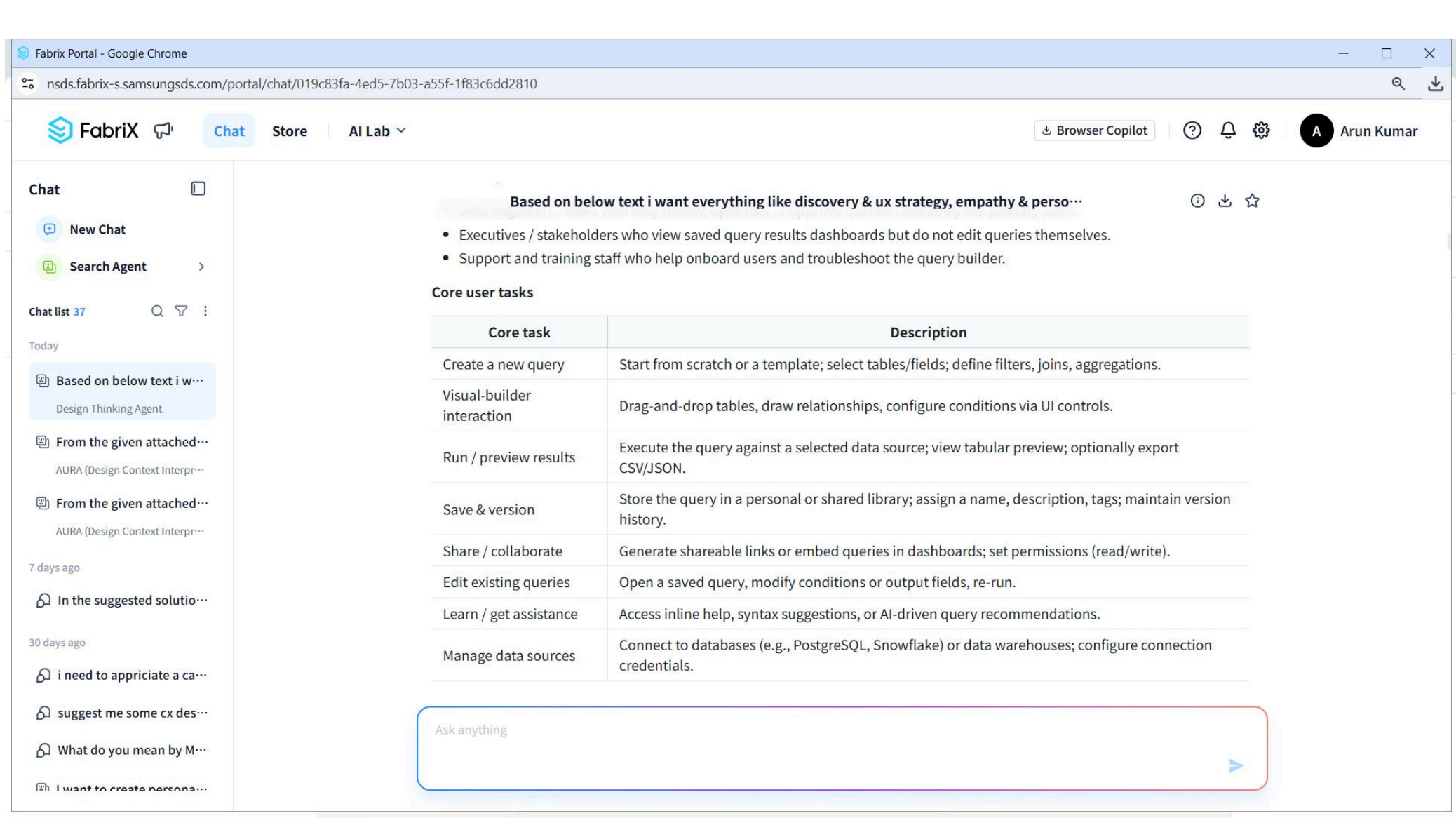
Platform & product type

- Web-based SaaS application (responsive for desktop+tablet; may also have a lightweight mobile view).
- Business-intelligence / data-query tool – "Quick Query Design" – enables users to compose, test, and share database queries (SQL or visual query builder) without deep technical knowledge.

Feature / screen intent

- Define the overall user-experience foundation for the Quick Query Design product.
- Primary intent is to understand who will use the query-design interface, how they will arrive at it, what tasks they need to accomplish, and how the product should be organised to support those tasks efficiently.

Primary product audience

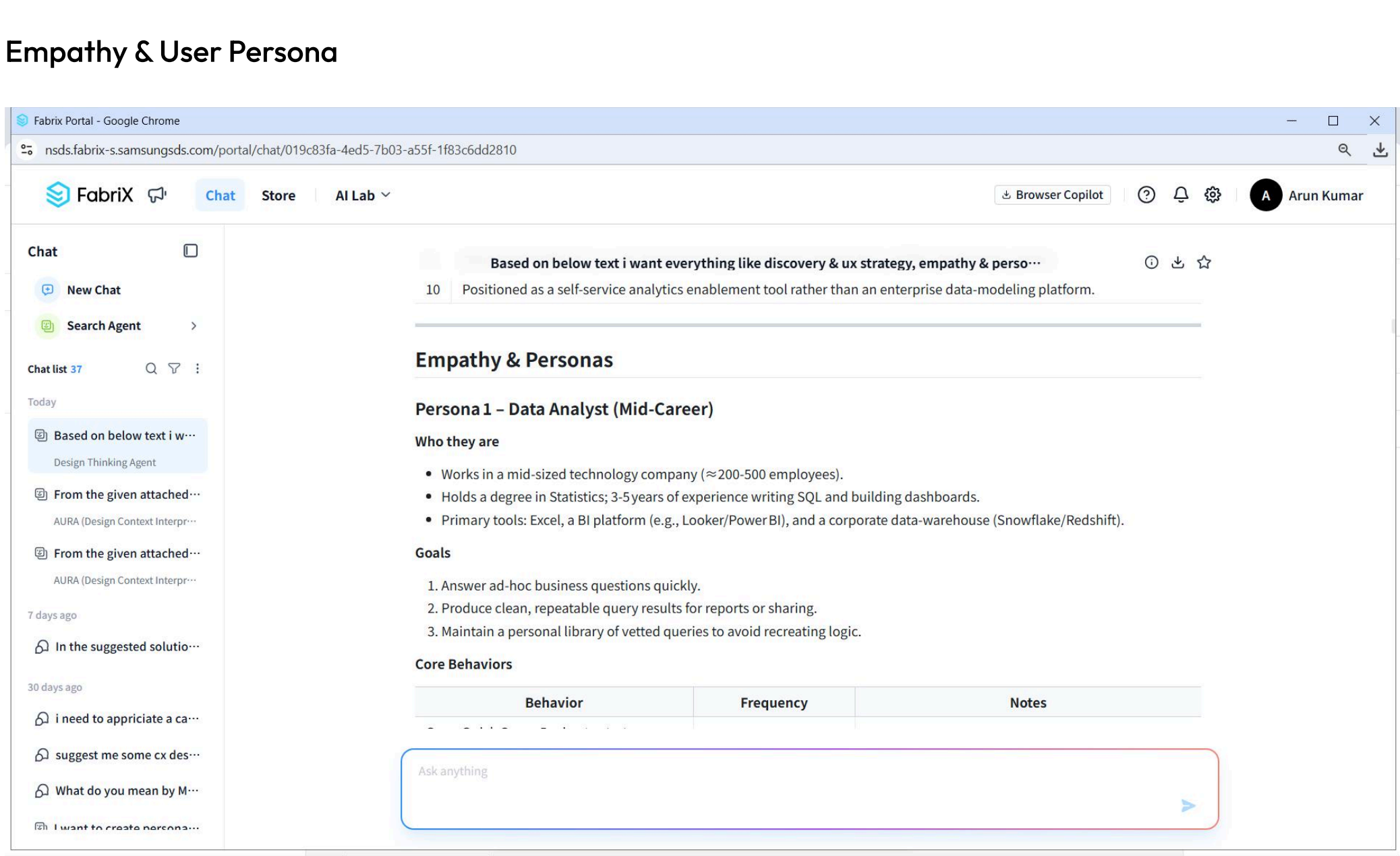


- Executives / stakeholders who view saved query results dashboards but do not edit queries themselves.
- Support and training staff who help onboard users and troubleshoot the query builder.

Core user tasks

Core task	Description
Create a new query	Start from scratch or a template; select tables/fields; define filters, joins, aggregations.
Visual builder interaction	Drag-and-drop tables, draw relationships, configure conditions via UI controls.
Run / preview results	Execute the query against a selected data source; view tabular preview; optionally export CSV/JSON.
Save & version	Store the query in a personal or shared library; assign a name, description, tags; maintain version history.
Share / collaborate	Generate shareable links or embed queries in dashboards; set permissions (read/write).
Edit existing queries	Open a saved query, modify conditions or output fields, re-run.
Learn / get assistance	Access inline help, syntax suggestions, or AI-driven query recommendations.
Manage data sources	Connect to databases (e.g., PostgreSQL, Snowflake) or data warehouses; configure connection credentials.

Empathy & User Persona



10 Positioned as a self-service analytics enablement tool rather than an enterprise data-modeling platform.

Empathy & Personas

Persona 1 – Data Analyst (Mid-Career)

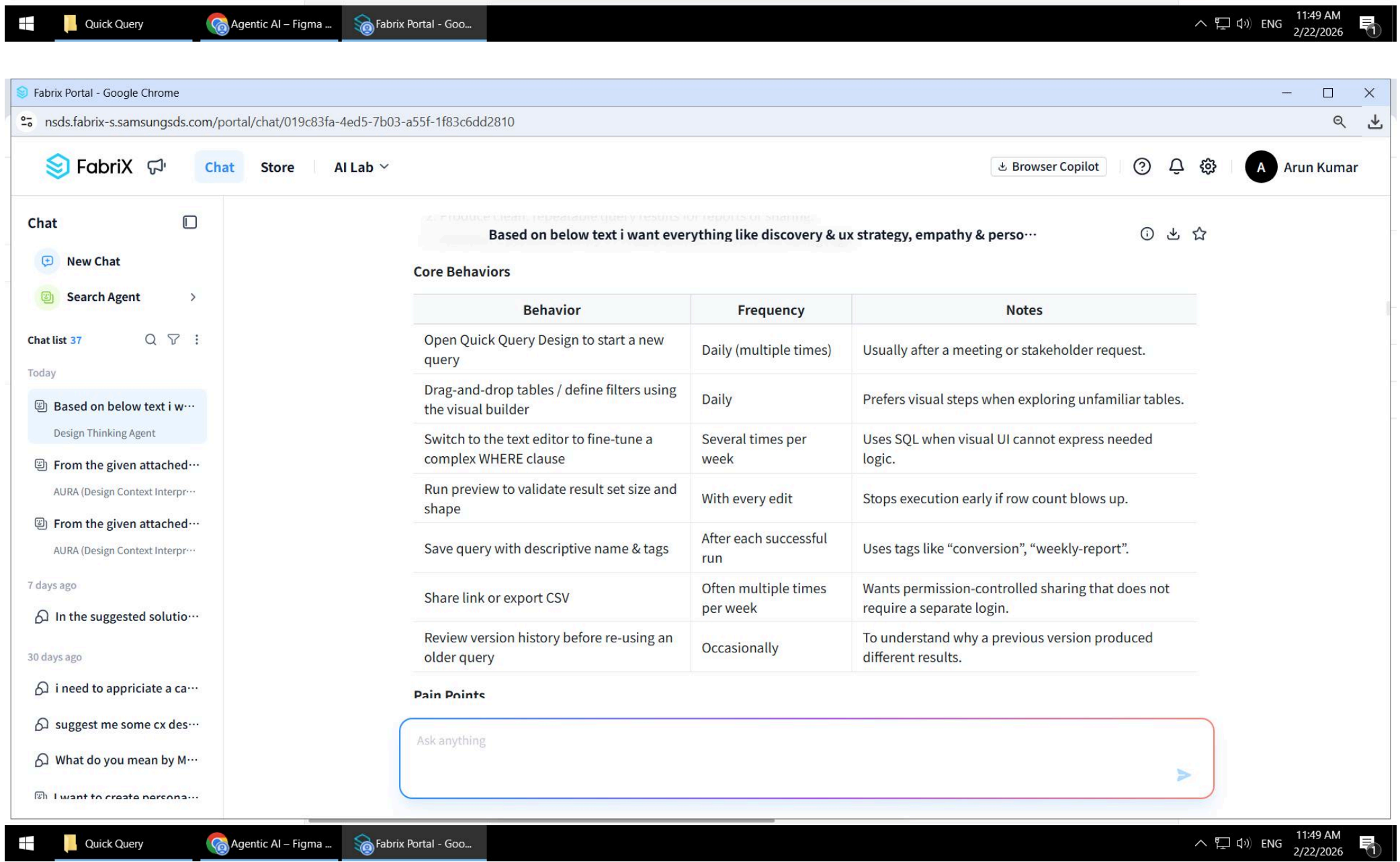
Who they are

- Works in a mid-sized technology company (≈200-500 employees).
- Holds a degree in Statistics; 3-5 years of experience writing SQL and building dashboards.
- Primary tools: Excel, a BI platform (e.g., Looker/PowerBI), and a corporate data warehouse (Snowflake/Redshift).

Goals

1. Answer ad-hoc business questions quickly.
2. Produce clean, repeatable query results for reports or sharing.
3. Maintain a personal library of vetted queries to avoid recreating logic.

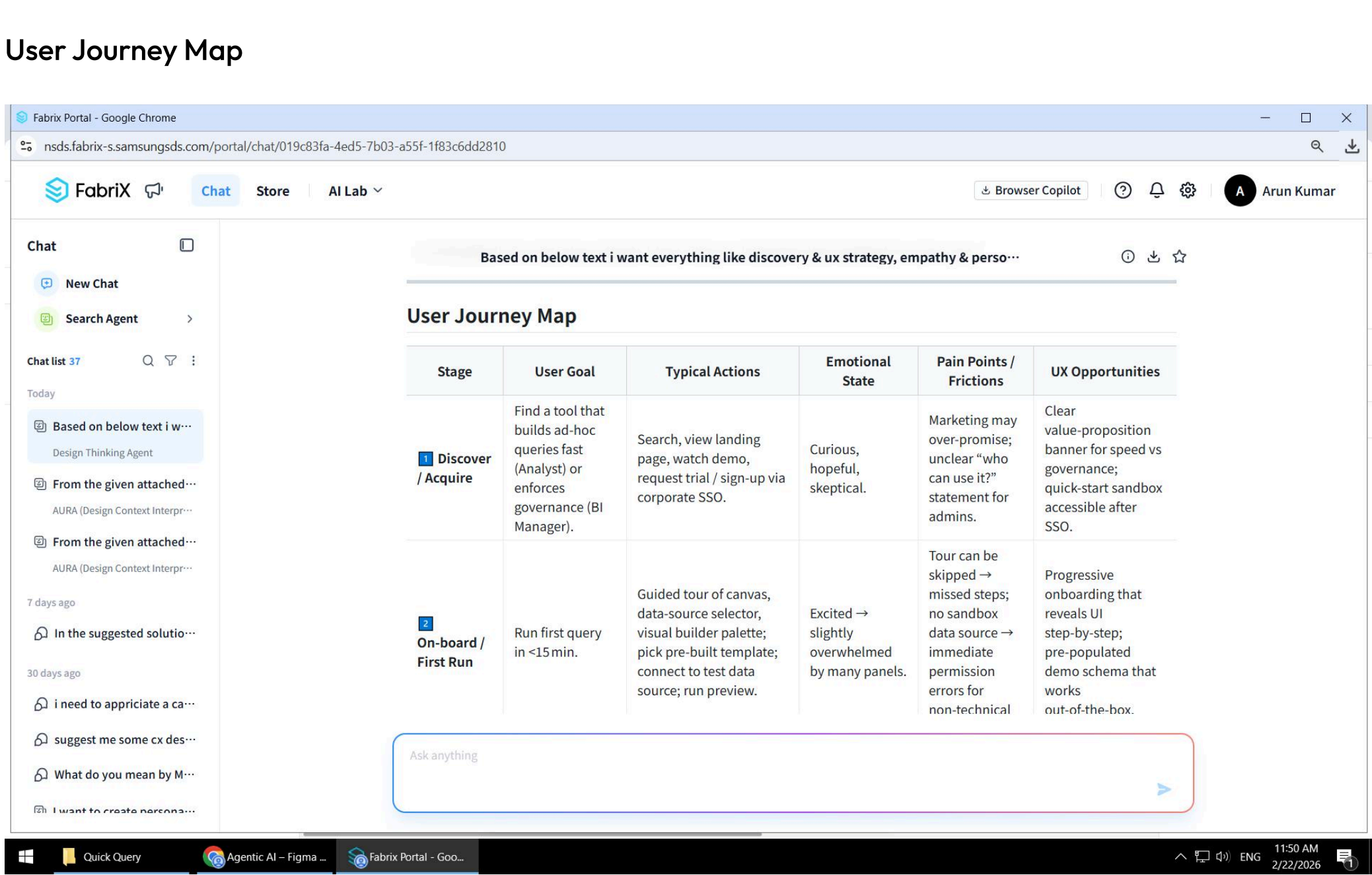
Core Behaviors



Behavior	Frequency	Notes
Open Quick Query Design to start a new query	Daily (multiple times)	Usually after a meeting or stakeholder request.
Drag-and-drop tables / define filters using the visual builder	Daily	Prefers visual steps when exploring unfamiliar tables.
Switch to the text editor to fine-tune a complex WHERE clause	Several times per week	Uses SQL when visual UI cannot express needed logic.
Run preview to validate result set size and shape	With every edit	Stops execution early if row count blows up.
Save query with descriptive name & tags	After each successful run	Uses tags like "conversion", "weekly-report".
Share link or export CSV	Often multiple times per week	Wants permission-controlled sharing that does not require a separate login.
Review version history before re-using an older query	Occasionally	To understand why a previous version produced different results.

Pain Points

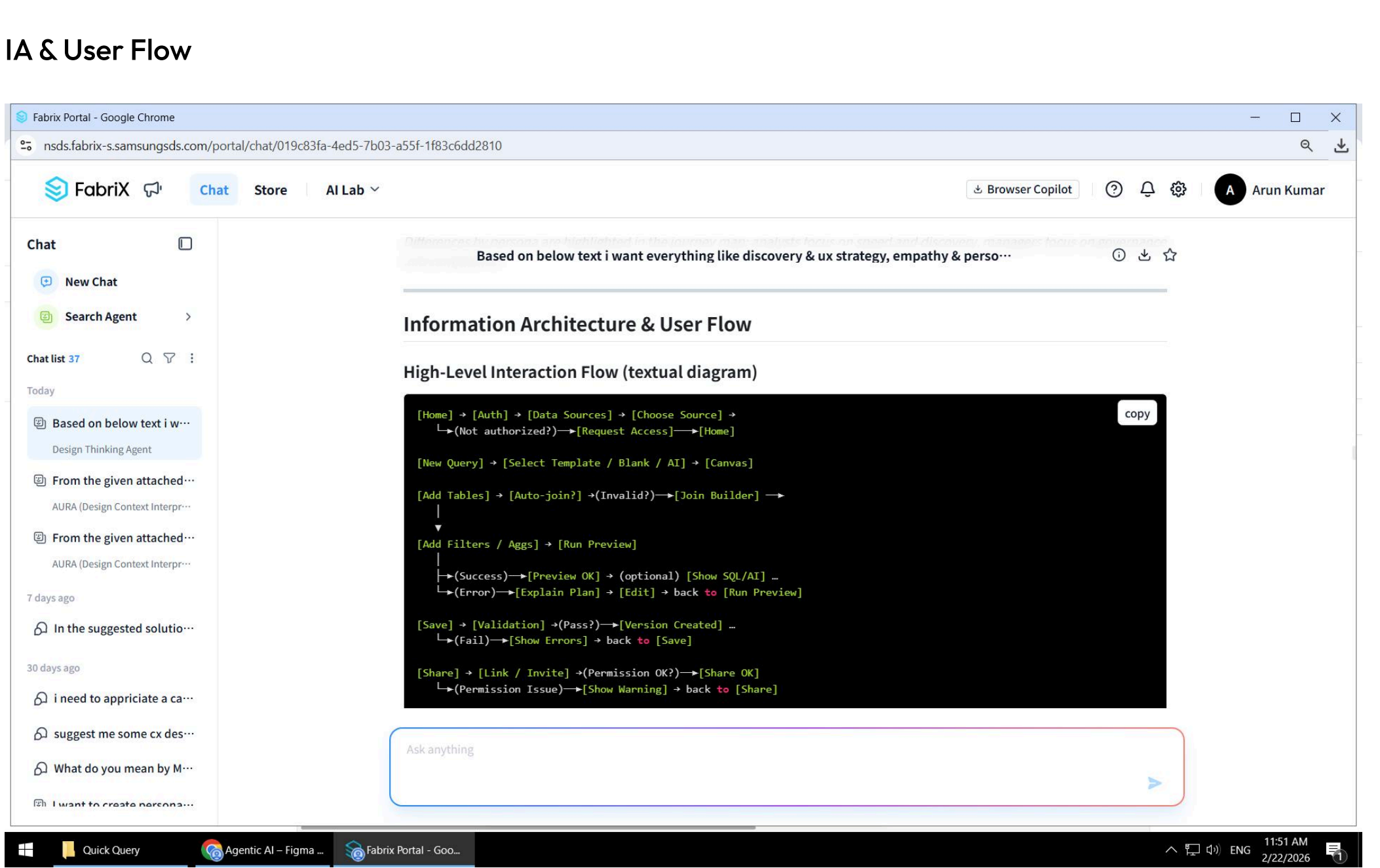
User Journey Map



User Journey Map

Stage	User Goal	Typical Actions	Emotional State	Pain Points / Frictions	UX Opportunities
1 Discover / Acquire	Find a tool that builds ad-hoc queries fast (Analyst) or enforces governance (BI Manager).	Search, view landing page, watch demo, request trial / sign-up via corporate SSO.	Curious, hopeful, skeptical.	Marketing may over-promise; unclear "who can use it?" statement for admins.	Clear value-proposition banner for speed vs governance; quick-start sandbox accessible after SSO.
2 On-board / First Run	Run first query in <15 min.	Guided tour of canvas, data-source selector, visual builder palette; pick pre-built template; connect to test data source; run preview.	Excited → slightly overwhelmed by many panels.	Tour can be skipped → missed steps; no sandbox data source → immediate permission errors for non-technical	Progressive onboarding that reveals UI step-by-step; pre-populated demo schema that works out-of-the-box.

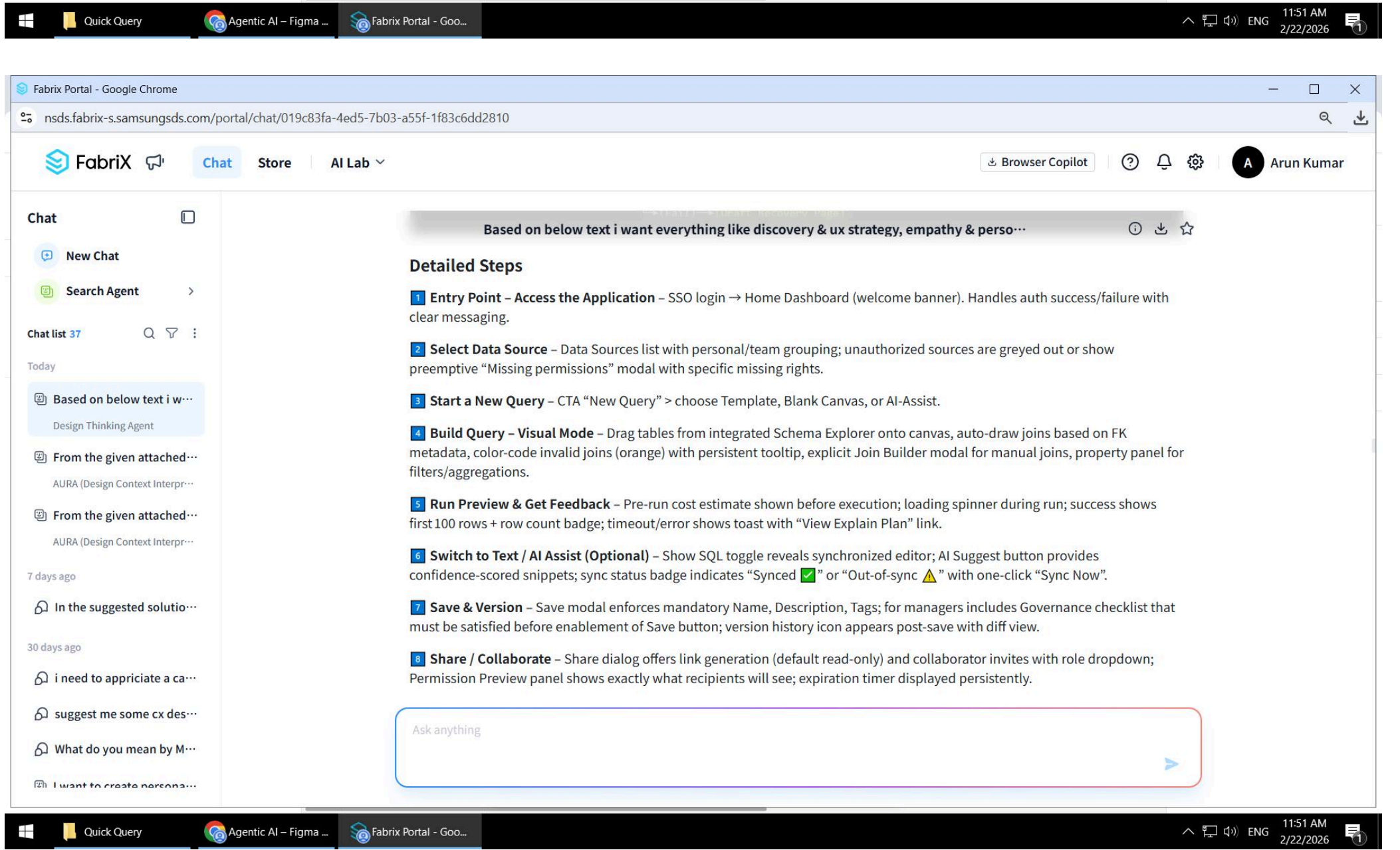
IA & User Flow



Information Architecture & User Flow

High-Level Interaction Flow (textual diagram)

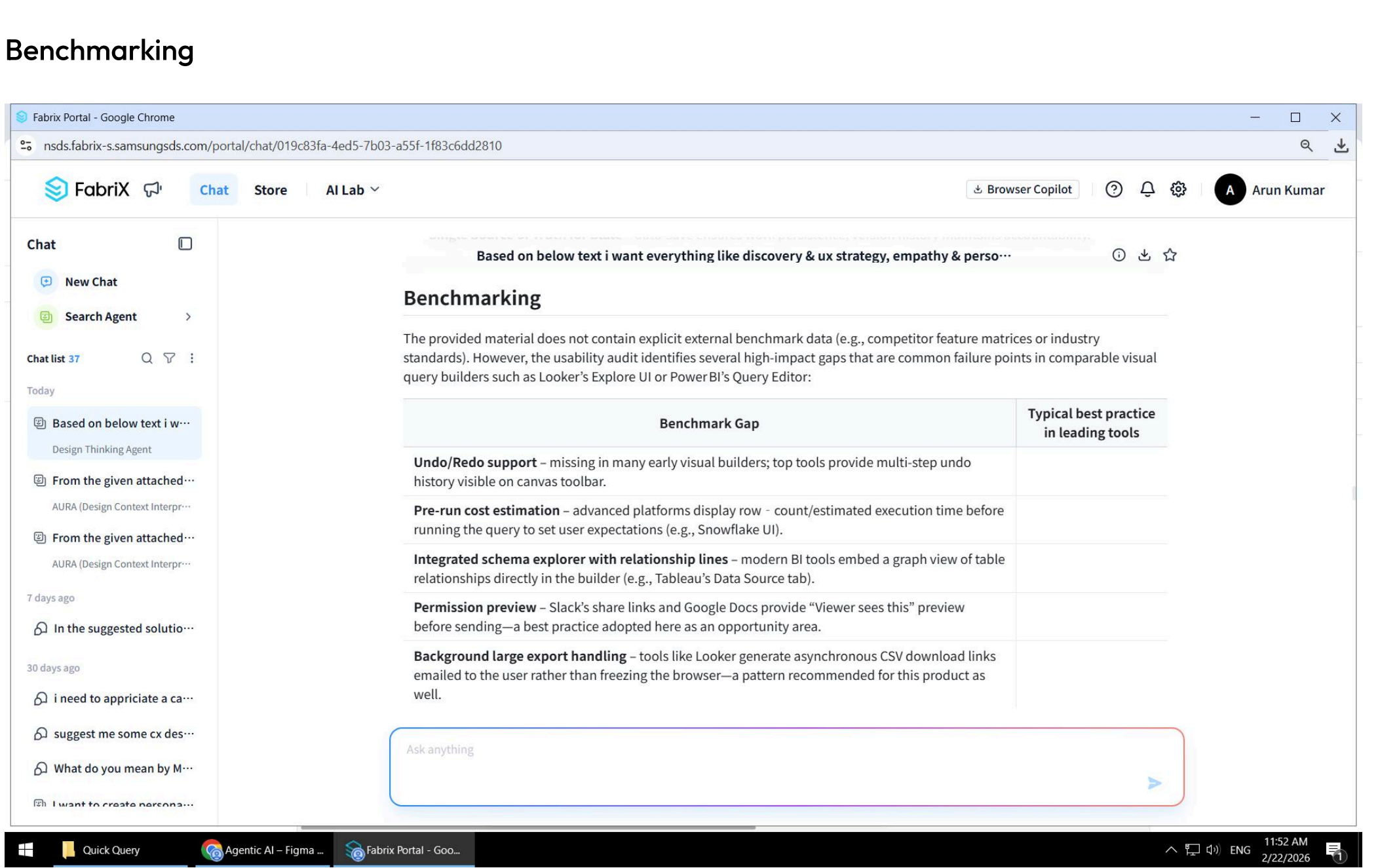
```
[Home] → [Auth] → [Data Sources] → [Choose Source] →
    └→[Not authorized?]→[Request Access]→[Home]

[New Query] → [Select Template / Blank / AI] → [Canvas]

[Add Tables] → [Auto-join?] →[Invalid?]→[Join Builder] →

[Add Filters / Aggs] → [Run Preview]
    |
    ├→[Success]→[Preview OK] → (optional) [Show SQL/AI] …
    └→[Error]→[Explain Plan] → [Edit] → back to [Run Preview]

[Save] → [Validation] →[Pass?]→[Version Created] …
    └→[Fail]→[Show Errors] → back to [Save]

[Share] → [Link / Invite] →[Permission OK?]→[Share OK]
    └→[Permission Issue]→[Show Warning] → back to [Share]
```



Detailed Steps

1. **Entry Point – Access the Application** – SSO login → Home Dashboard (welcome banner). Handles auth success/failure with clear messaging.
2. **Select Data Source** – Data Sources list with personal/team grouping; unauthorized sources are greyed out or show preemptive "Missing permissions" modal with specific missing rights.
3. **Start a New Query** – CTA "New Query" → choose Template, Blank Canvas, or AI-Assist.
4. **Build Query – Visual Mode** – Drag tables from integrated Schema Explorer onto canvas, auto-draw joins based on FK metadata, color-code invalid joins (orange) with persistent tooltip, explicit Join Builder modal for manual joins, property panel for filters/aggregations.
5. **Run Preview & Get Feedback** – Pre-run cost estimate shown before execution; loading spinner during run; success shows first 100 rows + row count badge; timeout/error shows toast with "View Explain Plan" link.
6. **Switch to Text / AI Assist (Optional)** – Show SQL toggle reveals synchronized editor; AI Suggest button provides confidence-scored snippets; sync status badge indicates "Synced ✅" or "Out-of-sync ⚠️" with one-click "Sync Now".
7. **Save & Version** – Save modal enforces mandatory Name, Description, Tags; for managers includes Governance checklist that must be satisfied before enablement of Save button; version history icon appears post-save with diff view.
8. **Share / Collaborate** – Share dialog offers link generation (default read-only) and collaborator invites with role dropdown; Permission Preview panel shows exactly what recipients will see; expiration timer displayed persistently.

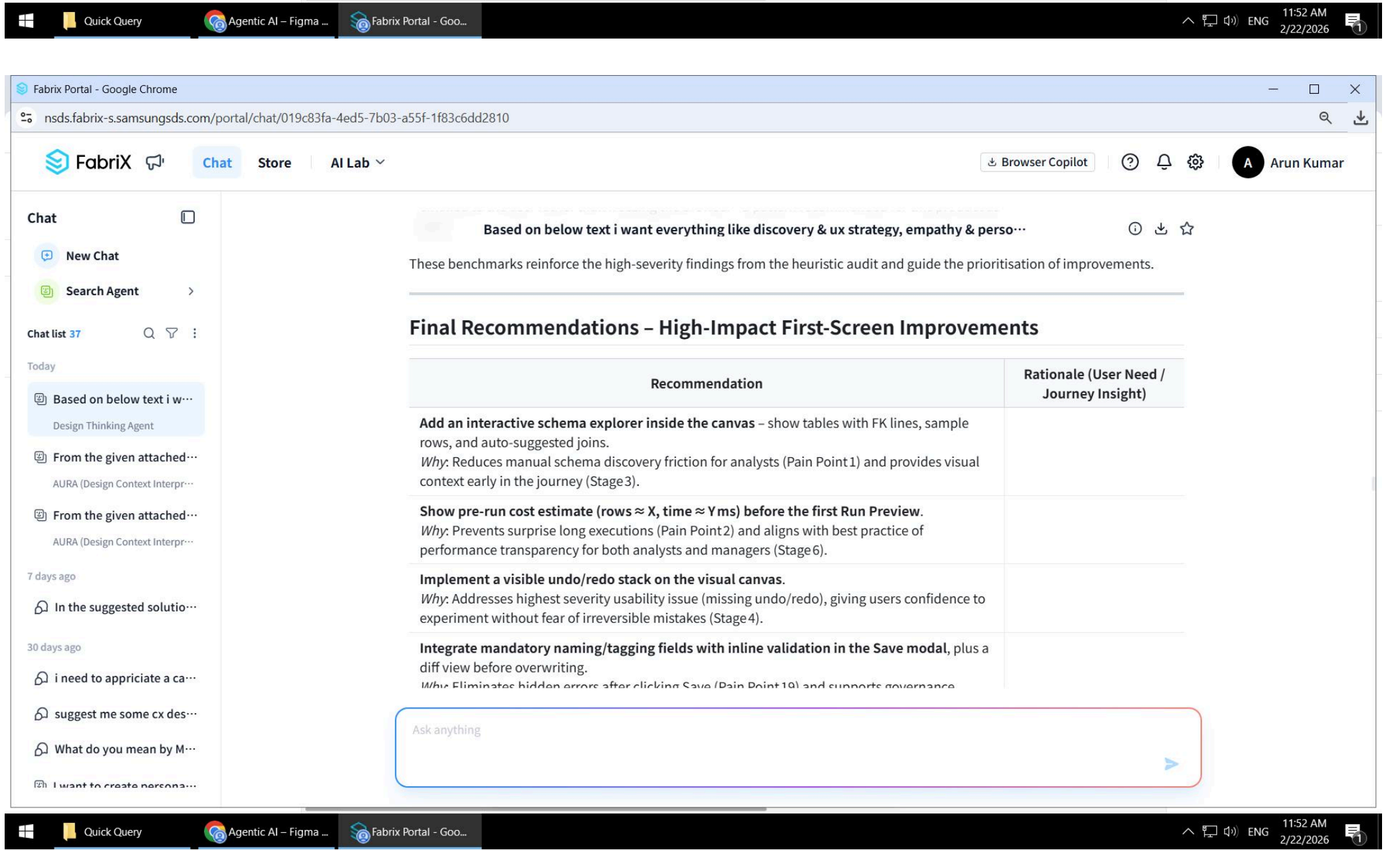
Benchmarking



Benchmarking

The provided material does not contain explicit external benchmark data (e.g., competitor feature matrices or industry standards). However, the usability audit identifies several high-impact gaps that are common failure points in comparable visual query builders such as Looker's Explore UI or Power BI's Query Editor:

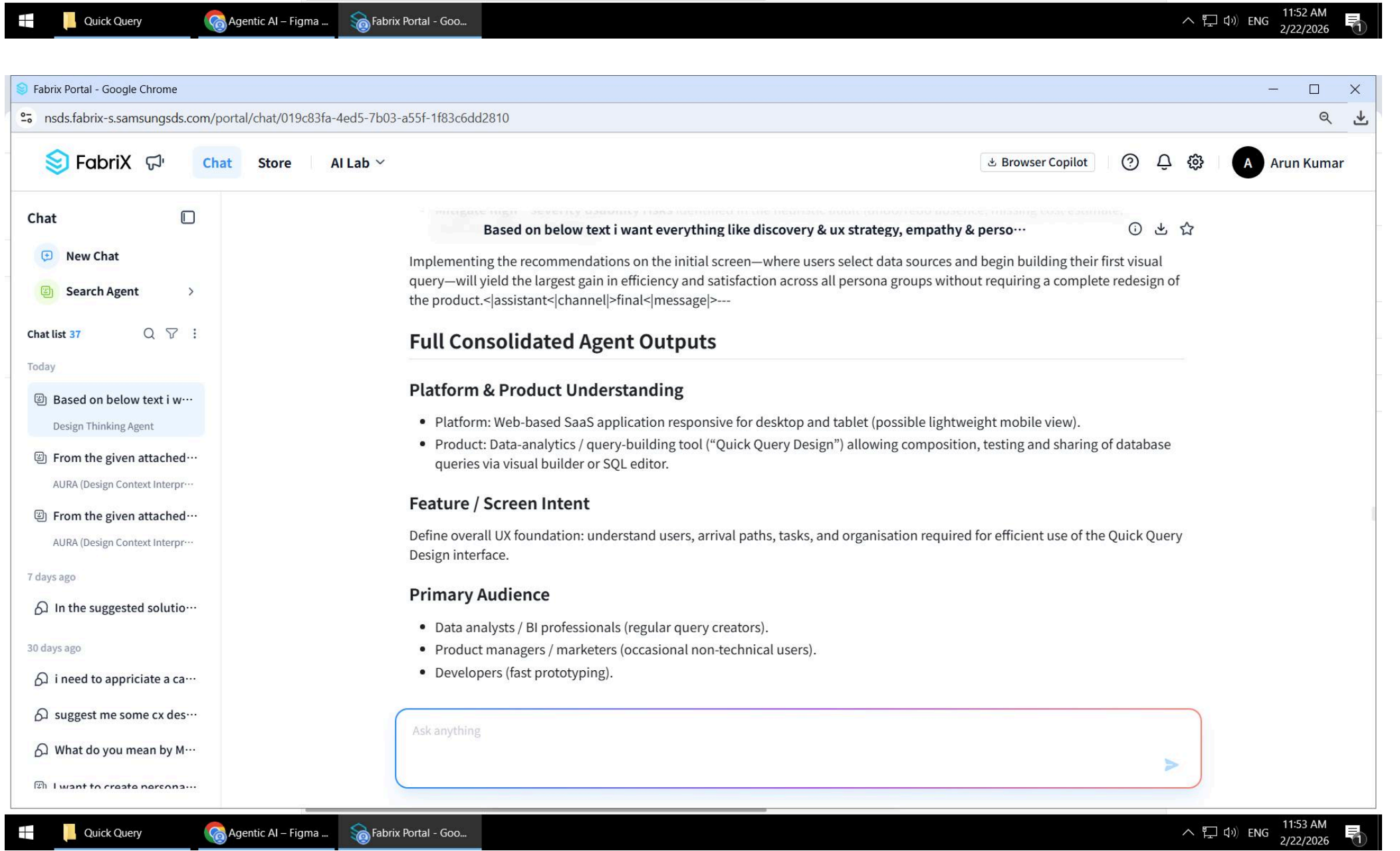
Benchmark Gap	Typical best practice in leading tools
Undo/Redo support – missing in many early visual builders; top tools provide multi-step undo history visible on canvas toolbar.	
Pre-run cost estimation – advanced platforms display row-count/estimated execution time before running the query to set user expectations (e.g., Snowflake UI).	
Integrated schema explorer with relationship lines – modern BI tools embed a graph view of table relationships directly in the builder (e.g., Tableau's Data Source tab).	
Permission preview – Slack's share links and Google Docs provide "Viewer sees this" preview before sending—a best practice adopted here as an opportunity area.	
Background large export handling – tools like Looker generate asynchronous CSV download links emailed to the user rather than freezing the browser—a pattern recommended for this product as well.	



These benchmarks reinforce the high-severity findings from the heuristic audit and guide the prioritisation of improvements.

Final Recommendations – High-Impact First-Screen Improvements

Recommendation	Rationale (User Need / Journey Insight)
Add an interactive schema explorer inside the canvas – show tables with FK lines, sample rows, and auto-suggested joins. *Why:* Reduces manual schema discovery friction for analysts (Pain Point 1) and provides visual context early in the journey (Stage 3).	
Show pre-run cost estimate (rows ≈ X, time ≈ Y ms) before the first Run Preview. *Why:* Prevents surprise long executions (Pain Point 2) and aligns with best practice of performance transparency for both analysts and managers (Stage 6).	
Implement a visible undo/redo stack on the visual canvas. *Why:* Addresses highest severity usability issue (missing undo/redo), giving users confidence to experiment without fear of irreversible mistakes (Stage 4).	
Integrate mandatory naming/tagging fields with inline validation in the Save modal, plus a diff view before overwriting.	



Implementing the recommendations on the initial screen—where users select data sources and begin building their first visual query—will yield the largest gain in efficiency and satisfaction across all persona groups without requiring a complete redesign of the product.<|assistant|channel|>final<|message|>—

Full Consolidated Agent Outputs

Platform & Product Understanding

- Platform: Web-based SaaS application responsive for desktop and tablet (possible lightweight mobile view).
- Product: Data analytics / query-building tool ("Quick Query Design") allowing composition, testing and sharing of database queries via visual builder or SQL editor.

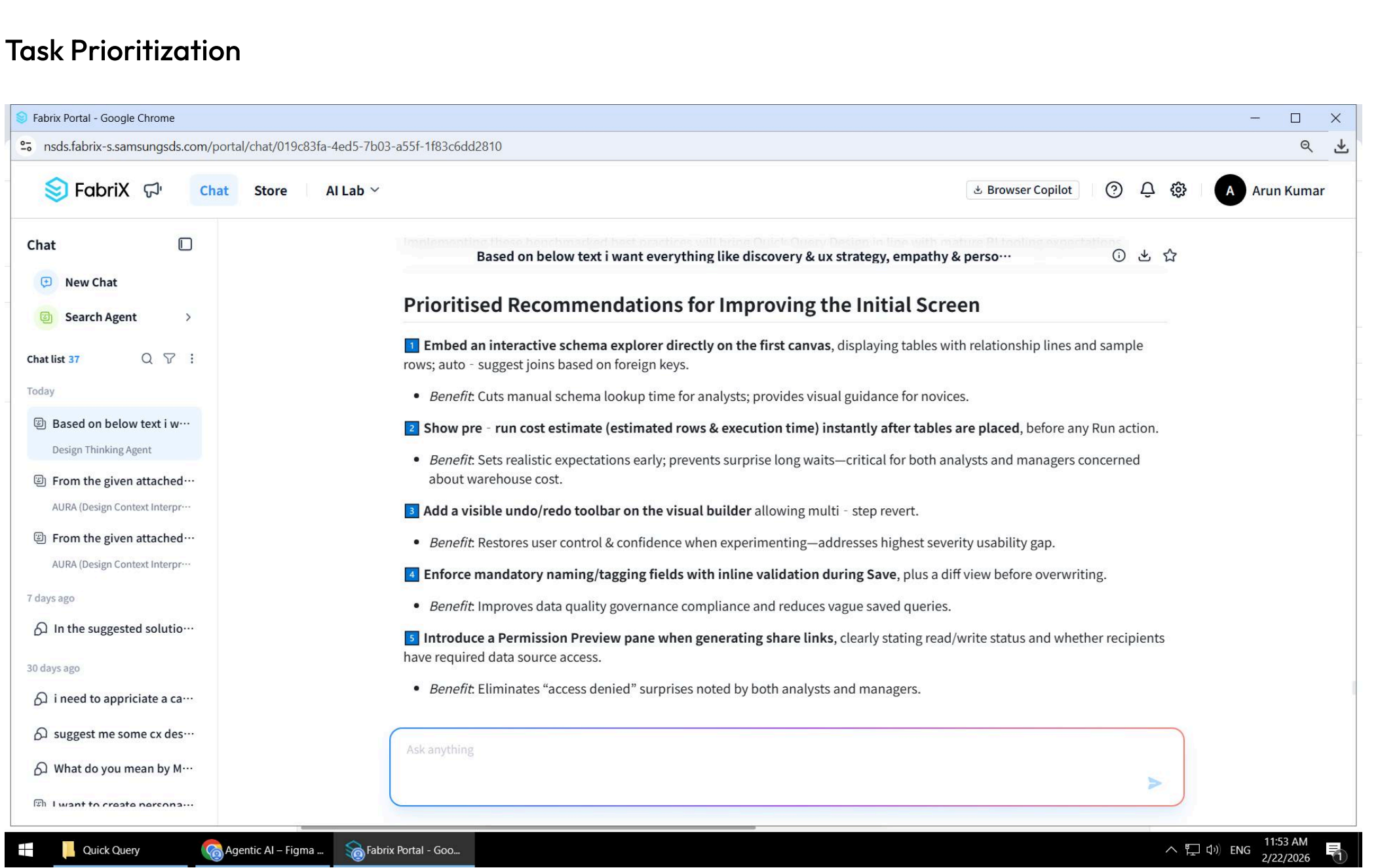
Feature / Screen Intent

Define overall UX foundation: understand users, arrival paths, tasks, and organisation required for efficient use of the Quick Query Design interface.

Primary Audience

- Data analysts / BI professionals (regular query creators).
- Product managers / marketers (occasional non-technical users).
- Developers (fast prototyping).

Task Prioritization



Prioritised Recommendations for Improving the Initial Screen

1. **Embed an interactive schema explorer directly on the first canvas**, displaying tables with relationship lines and sample rows; auto-suggest joins based on foreign keys.
   - *Benefit:* Cuts manual schema lookup time for analysts; provides visual guidance for novices.
2. **Show pre-run cost estimate (estimated rows & execution time) instantly after tables are placed**, before any Run action.
   - *Benefit:* Sets realistic expectations early; prevents surprise long waits—critical for both analysts and managers concerned about warehouse cost.
3. **Add a visible undo/redo toolbar on the visual builder** allowing multi-step revert.
   - *Benefit:* Restores user control & confidence when experimenting—addresses highest severity usability gap.
4. **Enforce mandatory naming/tagging fields with inline validation during Save**, plus a diff view before overwriting.
   - *Benefit:* Improves data quality governance compliance and reduces vague saved queries.
5. **Introduce a Permission Preview pane when generating share links**, clearly stating read/write status and whether recipients have required data source access.
   - *Benefit:* Eliminates "access denied" surprises noted by both analysts and managers.

Impact on Design Output

1. Faster requirement understanding
2. Better structured design artifacts
3. Improve resource efficiency & productivity
4. Improved design consistency
5. Higher confidence in decisions